

Joint Similar User Exploration and Informative Behavior Guidance for Multi-Modal New Item Recommendation

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Abstract

Multi-modal recommendation has become essential with the rapid expansion of online platforms such as e-commerce and video-sharing applications. In this work, we focus on the Multi-Modal New Item Recommendation (MMNIR) problem, where items with multi-modal content but newly introduced items lack interaction history. The MMNIR problem is particularly challenging in two aspects: (1) a large number of new items are created rapidly over time without any interaction data, (2) not all existing interactions are equally useful, and it is non-trivial to identify informative behaviors from users with similar preferences. However, previous methods fail to identify users with similar preferences and to capture informative behaviors from historical data. Furthermore, conventional models primarily rely on simple co-occurring signals, leading to

spurious neighbors and neglecting the informative behaviors of truly similar users with consistent preferences. To fill this gap, we propose Joint Similar User Exploration and Informative Behavior Guidance (SuperG) for solving the MMNIR problem. SuperG first proposes a similar user exploration module to identify users with similar preferences to the target user. Then it incorporates an informative behavior mining module to retrieve informative behaviors from both the target user and similar users' histories to support new item recommendation. Finally, SuperG proposes a behavior-guided hybrid recommendation module to incorporate the retrieved behavioral signals to guide the recommendation of new items. Our empirical study on three real datasets demonstrates that SuperG outperforms the state-of-the-art models under the MMNIR setting.

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CCS Concepts

• Information systems → Recommender systems.

Keywords

Recommender Systems, Multi-Modal Recommendation, New Item Recommendation, Large Language Models

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1 Introduction

With the rapid expansion of the Web, Multi-Modal Recommendation (MMR) models have become essential for online platforms such as e-commerce and video-sharing applications. These MMR models leverage both user-item interactions and multi-modal content features to improve recommendation accuracy [2, 12, 39, 64, 79]. However, in practical scenarios, these models face critical challenges: newly introduced items have no interaction data, which makes accurate recommendation difficult [7, 80, 85]. This issue becomes even more critical for these new items with multi-modal content, as both collaborative signals and modality preferences are unavailable [3, 44]. Thus, how to effectively handle cold-start items under multi-modal conditions still needs more investigation.

In this paper, we focus on the Multi-Modal New Item Recommendation (MMNIR) problem as shown in Fig. 1. That is, we aim to address the cold-start problem for newly introduced items with multi-modal content. For each target user, existing user-item interaction data is available, while new items lack any interaction history. Meanwhile, new items contain multi-modal content such as images, audio, and text. Moreover, we assume that certain behaviors of the target user as well as similar users can serve as informative signals. For instance, if the target user has previously interacted with items that are similar to the new item, such behaviors can inform the recommendation of the new item. Likewise, if similar users who share comparable preferences have interacted with items similar to the new item, such behaviors can also serve as informative signals to guide the recommendation. There are two main challenges in solving the MMNIR problem: (1) **CH1**: How to identify users who share similar preferences with the target user without explicit user similarity labels? (2) **CH2**: How to retrieve informative behaviors that can support recommendations for new items?

Previous methods cannot better solve MMNIR well. On the one hand, previous methods often focus primarily on modeling multi-modal content to learn user preferences [15, 18, 37, 63], while neglecting the valuable information contained in users' interaction history, which prevents previous approaches from performing effectively. Recent MMNIR models attempt to overcome this limitation by combining collaborative filtering (CF) [24, 82] signals and multi-modal information to better capture user preferences. However, they typically rely on simple co-occurring implicit signals and assume that users who have interacted with the same items are similar. This approach can lead to spurious neighbors, since users who interact with the same popular items may still have very different overall preferences, while truly similar users may never interact with the same items, causing their informative behaviors to be overlooked. Moreover, these methods often utilize Graph Neural Networks (GNNs) [41, 66, 68] to aggregate interaction information through user-item graph message passing. However, GNNs

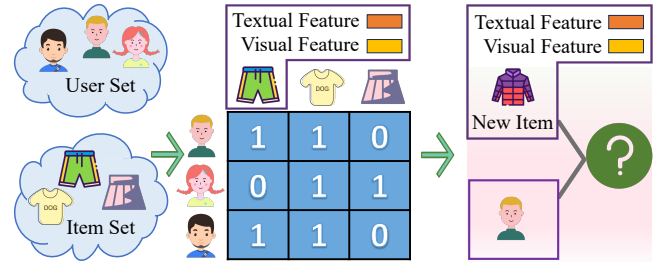


Figure 1: The problem definition of MMNIR.

struggle to capture explicit and semantically rich CF information and commonly suffer from over-smoothing [6, 32], which limits their ability to model comprehensive and discriminative interaction patterns. Although recent methods such as MMHCL [19] and MHCR [42] employ hypergraphs [14, 17, 70] to model higher-order user-item interactions, the adopted hypergraph-based architectures are sensitive to the choice of hyper-parameters and may fail to efficiently capture these higher-order relationships. Hence, they inevitably incur imprecise and meaningless hyperedges among user-item hypergraphs, leading to limited model potential. Therefore, traditional MMNIR models cannot solve **CH1** well. On the other hand, recent works leverage large language models (LLMs) [5, 13, 48, 55] to address MMNIR. However, these LLM-based models [62, 71] have not explicitly explored informative behaviors to guide the recommendation process. These methods often rely on a significant number of historical interactions, inevitably introducing noisy signals. Moreover, LLMs cannot simultaneously process all items in natural language form due to input length limitations, leading to substantial token consumption and making them impractical for real-world deployment [1]. To address this issue, the latest studies attempt to recall a subset of items or interactions as recommendation signals [28, 78]. Nevertheless, these approaches typically depend on simple text similarity and pay limited attention to the distinguishability of features among users and items, resulting in the retrieval of large but low-quality item sets that lack discriminative capacity and thus fail to provide truly informative signals for recommendation. Therefore, traditional MMNIR models cannot solve **CH2** well. In conclusion, previous methods cannot provide satisfactory results in tackling the MMNIR problem.

To address the aforementioned issues, in this paper, we propose Joint Similar User Exploration and Informative Behavior Guidance model (SuperG) for solving the MMNIR problem. SuperG includes three main modules, i.e., similar user exploration module, informative behavior mining module, and behavior-guided hybrid recommendation module. The similar user exploration module is designed to identify users most similar to the target user, introducing collaborative signals based on user-user similarity to address **CH1**. Specifically, we employ contrastive learning [29, 33, 54] to learn discriminative user representations for effective similar user exploration. In the informative behavior mining module, we retrieve and rerank the most informative user-item interactions from the target user and similar users' history to support the recommendation for solving **CH2**, as illustrated in Fig. 2. We refer to these interactions as informative behaviors. Finally, the behavior-guided hybrid recommendation module is set to leverage the informative

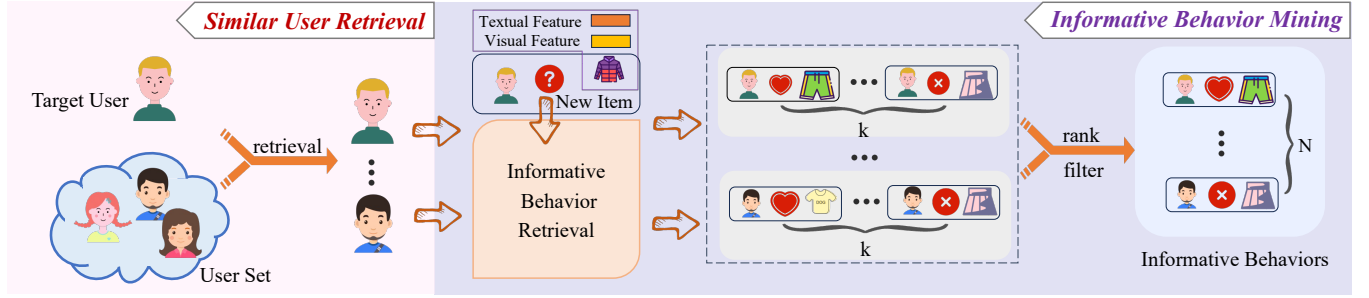


Figure 2: The illustration of similar user retrieval and informative behavior mining.

behaviors to guide the LLM in generating recommendations. In this module, we jointly utilize natural language and embedding-based representations to guide LLMs in predicting user-item interactions. By incorporating these three modules, SuperG can better identify users who share similar preferences with the target user without explicit user similarity labels, meanwhile SuperG can also identify and leverage the informative behaviors of the target user and similar users to alleviate the cold-start problem for new items.

We summarize our main contributions as follows: (1) We propose a novel framework, i.e., SuperG, for solving the MMNIR problem by combining similar user exploration, informative behavior mining, and LLM-based behavior-guided recommendation. (2) We propose similar users exploration via contrastive learning and informative behaviors mining through LLM-based feedback, leveraging collaborative signals to improve recommendation performance for cold-start items. (3) We introduce a behavior-guided hybrid recommendation approach that leverages informative behaviors in both textual and embedding forms to guide LLMs in recommending cold-start items. (4) We provide theoretical analyses and conduct extensive experiments to validate the effectiveness of SuperG.

2 Method

In this section, we will introduce the details of the proposed SuperG for solving the MMNIR problem. First, we describe notations. We assume there are two sets, i.e., a user set $\mathcal{U}$ and an item set $\mathcal{V}$. There exist N_u users and N_v items in the user and item sets respectively. $\mathbf{R} \in \mathbb{R}^{N_u \times N_v}$ denotes the user-item interaction matrix, where each entry $r_{ij} \in \{0, 1\}$ indicates whether user i has interacted with item j . Specifically, $r_{ij} = 1$ denotes an observed interaction and $r_{ij} = 0$ otherwise. Meanwhile, items in $\mathcal{V}$ have multi-modal features extracted from their content, such as visual and textual features. We adopt generic feature extractors to generate the multi-modal features. For an item $j \in \mathcal{V}$, we denote its feature representation as $\mathbf{x}_j \in \mathbb{R}^{M \times d}$, where M is the number of modalities and d is the dimensionality of each modality-specific feature vector.

The framework of SuperG is shown in Fig. 3. SuperG mainly has three modules, i.e., similar user exploration module, informative behavior mining module, and behavior-guided hybrid recommendation module. The similar user exploration module aims to identify users with similar preferences to the target user, thereby introducing collaborative signals. The informative behavior mining module is set to retrieve interaction behaviors from both the target user and similar users that are most likely to support the recommendation of new items through LLM-based feedback. The behavior-guided

hybrid recommendation module aims to incorporate the retrieved behavioral signals to guide the LLM in generating recommendations for new items. By incorporating these three modules in SuperG, we can leverage rich information from similar users and informative behaviors to enhance the recommendation of new multi-modal items. We will introduce these three modules in detail later.

2.1 Similar User Exploration Module

To start with, we first introduce the similar user exploration module in SuperG. The similar user exploration module includes two main steps, i.e., user representation construction and user representation contrastive learning. User representation construction is set to construct user embeddings from historical interactions. User representation contrastive learning aims to learn discriminative user representations for similar user retrieval. By combining user representation construction and user representation contrastive learning, the similar user exploration module can effectively identify users with similar preferences to the target user, thereby introducing collaborative signals to enhance new item recommendation.

User Representation Construction. Specifically, we represent a user's historical interaction sequence $\mathcal{H}_u$ as a collection of previously interacted items, each represented by the corresponding vector $\mathbf{z}_j$. We obtain $\mathbf{z}_j$ for each item by following [3, 61], and the details are presented in the Appendix B. These item vectors form a matrix $\mathbf{X}_u = [\mathbf{z}_1, \mathbf{z}_2, \dots, \mathbf{z}_N]^T \in \mathbb{R}^{N \times d}$, where N denotes the number of historical items and d is the embedding dimension. To capture the order information in the user's interaction history, we add positional embeddings $\mathbf{P} \in \mathbb{R}^{N \times d}$ to obtain $\tilde{\mathbf{X}}_u = \mathbf{X}_u + \mathbf{P}$. We then apply a transformer [57] encoder to $\tilde{\mathbf{X}}_u$ to model complex dependencies among the items in the sequence. The final user embedding is computed by averaging the transformer's output:

$$\mathbf{u} = \text{MEAN}(\text{Trm}(\tilde{\mathbf{X}}_u)) \in \mathbb{R}^d, \quad (1)$$

where $\text{Trm}(\cdot)$ denotes the transformer encoder.

User Representation Contrastive Learning. Although user representation construction has provided user embeddings derived from historical interactions, there are no explicit labels indicating which users are similar to each other. Therefore, it is essential to further explore underlying user-user similarities beyond individual representations. That is, we should learn more discriminative user representations to better identify users with similar preferences via contrastive learning. Hence, we propose the user representation contrastive learning method to achieve this goal. Specifically,

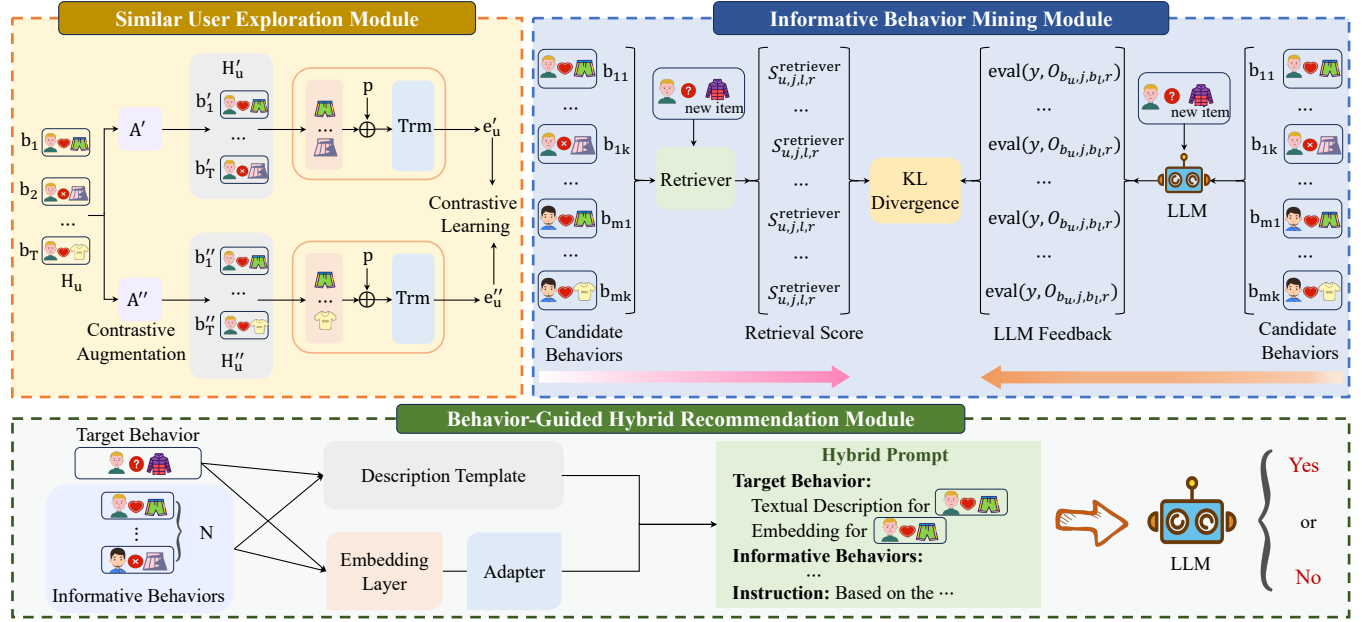


Figure 3: The framework of our proposed SuperG.

we first apply four contrastive augmentation methods to the user history $\mathcal{H}_u$ to obtain different views of the user's history:

Attention Masking. We randomly select a number of items from the user history and mask them by applying attention masks during the transformer encoding process (Eq.(1)).

Subsequence Sampling. We randomly select a continuous subsequence from the original user history $\mathcal{H}_u$.

Subsequence Reordering. A continuous subsequence is randomly selected and the order of items within this sub-sequence is shuffled.

Segment Shuffling. The user history is divided into multiple non-overlapping continuous segments. The order of these segments is then shuffled while preserving the internal item order.

Then, we randomly select two contrastive augmentation methods C' and C'' to generate two distinct views of the user's history $\mathcal{H}_u$, denoted as $\mathcal{H}'_u$ and $\mathcal{H}''_u$, respectively. These augmented histories are passed through the transformer encoder (as defined in Eq.(1)) to obtain the corresponding user embeddings $\mathbf{e}'_u$ and $\mathbf{e}''_u$. Since both embeddings are derived from different augmentations of the same user history, they are expected to preserve consistent user characteristics and are treated as a positive pair. In contrast, embeddings generated from other users within the same mini-batch are used as negative samples. We apply the InfoNCE loss [27, 38, 47, 73] to encourage the model to minimize the distance between the positive pairs while maximizing the distance to the negatives. The contrastive loss is formulated as:

$$\mathcal{L}_{CL} = - \left[\log \frac{\exp(\cos(\mathbf{e}'_u, \mathbf{e}''_u)/\tau_1)}{\sum_{u^- \in \mathcal{U}_{neg}} \exp(\cos(\mathbf{e}'_u, \mathbf{e}''_{u^-})/\tau_1)} + \log \frac{\exp(\cos(\mathbf{e}'_{u^-}, \mathbf{e}''_u)/\tau_1)}{\sum_{u^- \in \mathcal{U}_{neg}} \exp(\cos(\mathbf{e}'_{u^-}, \mathbf{e}''_u)/\tau_1)} \right], \quad (2)$$

where τ_1 is the temperature coefficient controlling the sharpness of the distribution, $\cos(\cdot)$ denotes the cosine similarity function, and $\mathcal{U}_{neg}$ is the set of randomly sampled negative users.

After training via Eq.(2), we obtain the user embedding $\mathbf{e}_u$ from Eq.(1). We then compute the cosine similarity between each pair of user embeddings, and retrieve the top- m most similar users for each user u , as illustrated in Fig. 2.

2.2 Informative Behavior Mining Module

After obtaining the top- m similar users, we propose an informative behavior mining module to identify the most informative behaviors from both the target user and each similar user's history to support the recommendation of new items. Specifically, for the target user u and a new item j , we retrieve the top- k items from the interaction histories of the target user u and m similar users to obtain a total of $(m+1) \times k$ candidate behaviors denoted as:

$$\mathcal{B}^{\text{retrieved}} = \{b_{l,r} \mid l \in \{1, \dots, m+1\}, r \in \{1, \dots, k\}\}, \quad (3)$$

where $b_{l,r}$ represents an interaction between user l and the historical item r . Specifically, we first define $b_{u,j}$ as the target behavior comprising user u and the new item j , and compute the semantic relevance between $b_{u,j}$ and each candidate behavior $b_{l,r}$ as follows:

$$S_{b_{u,j}, b_{l,r}}^{\text{retriever}} = \cos(\mathbf{e}_u, \mathbf{e}_l) + \cos(\mathbf{z}_j, \mathbf{z}_r), \quad (4)$$

where $\mathbf{e}_u$ and $\mathbf{e}_l$ denote the embeddings of the target user and user l , and $\mathbf{z}_j$ and $\mathbf{z}_r$ denote the embeddings of the target item j and candidate item r , respectively. Previous retrieval methods typically use $S_{b_{u,j}, b_{l,r}}^{\text{retriever}}$ directly for retrieval. However, the term $\cos(\mathbf{e}_u, \mathbf{e}_l)$ is shared across all behaviors from the similar user l , resulting in $S_{b_{u,j}, b_{l,r}}^{\text{retriever}}$ solely focusing on semantic relevance between items. Since different users might input the same query but expect different outputs due to their varying preferences, it is essential to consider user-specific preferences. For instance, suppose both *User A* and

User B have User C as a similar user. If we rely solely on item j to retrieve from User C's interaction history, both User A and User B would obtain the same candidate items, failing to reflect their individual preferences. To tackle this, we incorporate user personalization by computing a user-specific preference score for each candidate item:

$$S_{u,r}^{\text{retriever}} = \cos(\text{MLP}(\mathbf{e}_u), \mathbf{z}_r), \quad (5)$$

where MLP is a multi-layer perceptron [46] that maps the user embedding $\mathbf{e}_u$ to the space where the cosine similarity is computed. $\mathbf{e}_u$ is obtained via Eq.(1). The final retrieval score is a weighted combination of the semantic relevance score $S_{b_{u,j},b_{l,r}}^{\text{retriever}}$ and the user-specific preference score $S_{u,r}^{\text{retriever}}$:

$$S_{u,j,l,r}^{\text{retriever}} = (1 - \lambda)S_{b_{u,j},b_{l,r}}^{\text{retriever}} + \lambda S_{u,r}^{\text{retriever}}, \quad (6)$$

where λ balances semantic relevance and personalization.

Optimization. Although informative behavior retrieval can retrieve behaviors that reflect both semantic relevance and user-specific preferences, the retrieved results are not guaranteed to guide the LLM to generate responses that better match the target output y [40, 52]. Since there is no ground truth indicating which retrieved behaviors are better, we instead evaluate the alignment between the LLM's output and the target output y , using this as a label to optimize the informative behavior retrieval process. Specifically, we first retrieve the $(m + 1) \times k$ candidate behaviors based on $S_{b_{u,j},b_{l,r}}^{\text{retriever}}$ defined in Eq.(4). Then, we concatenate each candidate behavior with the target behavior $b_{u,j}$ to construct prompts for the LLM, resulting in $(m + 1) \times k$ outputs:

$$\begin{aligned} \{O_{b_{u,j},b_{l,r}} = \text{LLM}(b_{u,j}, b_{l,r}) \mid \\ l \in \{1, \dots, m + 1\}, \\ r \in \{1, \dots, k\}\}, \end{aligned} \quad (7)$$

where $\text{LLM}(b_{u,j}, b_{l,r})$ represents the output generated by inputting the prompt into the LLM. To ensure consistency between behavior retrieval and final recommendation prediction, we adopt a unified prompt format across both modules. The specific prompt design is detailed in the behavior-guided generative recommendation module. Based on the quality of these outputs, we can calculate the distribution over the candidate behaviors as follows:

$$p_{\text{LLM}}(b_{l,r} \mid b_{u,j}, y) = \frac{\exp(\text{eval}(y, O_{b_{u,j},b_{l,r}}))}{\sum_{l=1}^{m+1} \sum_{r=1}^k \exp(\text{eval}(y, O_{b_{u,j},b_{l,r}}))}, \quad (8)$$

where $\text{eval}(\cdot)$ measures the alignment between the target output y and the LLM's output $O_{b_{u,j},b_{l,r}}$. In our implementation, the LLM is prompted to generate a single-token response, either "Yes" or "No". We define the $\text{eval}(\cdot)$ as the log-probability of the correct token y under the output distribution of the LLM:

$$\text{eval}(y, O_{b_{u,j},b_{l,r}}) = \log p_{\text{LLM}}(y \mid b_{u,j}, b_{l,r}). \quad (9)$$

A larger value returned by $\text{eval}(\cdot)$ indicates a better generated result. Similarly, we compute the score distribution over the candidate

behaviors based on the retrieval score $S_{u,j,l,r}^{\text{retriever}}$ in Eq.(6):

$$p_{\text{retriever}}(b_{l,r} \mid b_{u,j}) = \frac{\exp(S_{u,j,l,r}^{\text{retriever}})}{\sum_{l=1}^{m+1} \sum_{r=1}^k \exp(S_{u,j,l,r}^{\text{retriever}})}. \quad (10)$$

We aim to retrieve behaviors that lead to better LLM-generated results, which means making the distribution $p_{\text{retriever}}(b_{l,r} \mid b_{u,j})$ in Eq.(10) closer to the distribution $p_{\text{LLM}}(b_{l,r} \mid b_{u,j}, y)$ in Eq.(8). To achieve this, we compute the KL divergence [11, 26, 56] between the two distributions as the loss to optimize the informative behavior retrieval process:

$$\mathcal{L}_{\text{retriever}} = \text{KL}(p_{\text{retriever}}(b_{l,r} \mid b_{u,j}) \parallel p_{\text{LLM}}(b_{l,r} \mid b_{u,j}, y)). \quad (11)$$

After we retrieve $\mathcal{B}_{\text{retrieved}}$ through the informative behavior retrieval based on $S_{u,j,l,r}^{\text{retriever}}$, we further refine the results by ranking $\mathcal{B}_{\text{retrieved}}$ to obtain the final top- N ranked results $\mathcal{B}_{\text{ranked}} = \{b_i \mid i \in \{1, \dots, N\}\}$ based on the retrieval score. Specifically, we sort all candidate behaviors $b_{l,r} \in \mathcal{B}_{\text{retrieved}}$ in descending order of their corresponding $S_{u,j,l,r}^{\text{retriever}}$ values, and select the top- N behaviors with the highest scores [51, 53]. This step ensures that the most semantically relevant and user-personalized behaviors are selected to guide the LLM in generating personalized recommendations.

2.3 Behavior-Guided Hybrid Recommendation Module

After obtaining the informative behaviors $\mathcal{B}_{\text{retrieved}}$ and their corresponding representations, we propose the behavior-guided hybrid recommendation module to leverage the informative behaviors to guide the LLM in generating personalized interaction predictions for the target user u and new item j . This module includes three main steps, i.e., prompt construction, LLM-compatible representation adaptation, and interaction prediction. Prompt construction is set to concatenate the target user, the new item, and the retrieved informative behaviors into a structured prompt. LLM-compatible representation adaptation aims to transform external embeddings into LLM-compatible representations [31, 50, 72] to integrate behavior-guided signals into LLMs. Interaction prediction is set to prompt the LLM to generate a binary response indicating whether the target user would interact with the new item.

Prompt Construction. We first convert the informative behaviors into textual descriptions in the following form:

Behavior _{i} : User l interacted with Item r ? "Yes"/"No"

Here, both users and items are represented by their IDs. We then append textual attributes or descriptions of items and users to the prompt when available. For example, the item description may include its title, category, or other relevant metadata. In addition to textual descriptions, we incorporate the latent representations (embeddings) of the user and item into the prompt, inspired by the approach in XRec [43]. These embeddings provide dense semantic information [36] that complements the textual features and enhances the model's understanding of user preferences and item characteristics. Additionally, we prepend an instruction to guide the LLM to generate the desired output. This instruction clearly defines the prediction task, which is to determine whether the target user would interact with the new item, and specifies the expected output

format, typically a binary response ("Yes" or "No"). As shown in Fig. 3, the complete prompt combines the instruction with the target user and new item information, as well as informative behaviors. The target user, the new item, and the informative behaviors all include textual descriptions and embeddings. The detailed instruction and full prompt design are introduced in the Appendix C.

LLM-Compatible Representation Adaptation. Since the user and item embeddings $\mathbf{e}_u$ and $\mathbf{z}_j$ originate from an external representation space, they may not align with the LLM's internal embedding distribution [21]. To bridge this gap, we introduce a lightweight adapter module $f(\cdot)$ that transforms raw embeddings into LLM-compatible representations. Formally:

$$\tilde{\mathbf{e}}_u = f(\mathbf{e}_u), \quad \tilde{\mathbf{z}}_j = f(\mathbf{z}_j). \quad (12)$$

The projected vectors $\tilde{\mathbf{e}}_u$ and $\tilde{\mathbf{z}}_j$ are then injected into the LLM prompt using special tokens:

$$[\text{USER_EMBED}] = \tilde{\mathbf{e}}_u, \quad [\text{ITEM_EMBED}] = \tilde{\mathbf{z}}_j. \quad (13)$$

Training. During training, we freeze all LLM parameters and optimize only the shared adapter module $f(\cdot)$. In this work, the adapter is built with a two-layer bottleneck structure (Linear–ReLU–Linear). The objective is to align the injected embeddings with the LLM's token-level reasoning, improving the prompt's semantic integrity. The training signal comes from the LLM's generative objective, which predicts a correct binary label ("Yes" or "No"). The adapter is trained end-to-end via the following loss:

$$\mathcal{L}_{\text{ad}} = -\frac{1}{N} \sum_{i=1}^N [y_i \log(\hat{y}_i) + (1 - y_i) \log(1 - \hat{y}_i)], \quad (14)$$

where $\hat{y}_i$ is the predicted probability of "Yes" generated by the LLM, and y_i is the ground truth label.

Joint Optimization with Behavior Retriever. To ensure consistency between behavior retrieval and final recommendation prediction, we share the adapter module $f(\cdot)$ across both the informative behavior mining module and the behavior-guided hybrid recommendation module. In retrieval, adapted embeddings $\tilde{\mathbf{e}}_u$ and $\tilde{\mathbf{z}}_j$ are used to construct prompts for retrieving informative behaviors. The adapter is jointly optimized by Eq.(11) and Eq.(14).

Interaction Prediction. Given the constructed prompt with textual descriptions and adapted embeddings, the LLM is prompted to directly generate a binary response indicating whether the target user u would interact with the new item j . The output is constrained to be one of the two tokens: "Yes" or "No", following the predefined output format. This prediction corresponds to the final recommendation result in the MMNIR task, effectively completing the recommendation process for the target user and new item pair.

2.4 Putting Together

The total loss of SuperG could be obtained by combining the losses from different modules:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{CL}} + \alpha \mathcal{L}_{\text{retriever}} + \beta \mathcal{L}_{\text{ad}}, \quad (15)$$

where α and β are hyper-parameters controlling the relative importance of each loss term. By doing this, SuperG can effectively leverage useful knowledge from existing user-item interactions to

alleviate the cold-start problem. Additionally, the inference algorithm of SuperG can be found in the Appendix D.

3 Experiments

In this section, we conduct experiments on several real-world datasets to answer the following questions: (1) **RQ1**: How does the proposed model **SuperG** perform compared with the state-of-the-art MMNIR methods? (2) **RQ2**: How do the similar user exploration and informative behavior guidance contribute to performance improvement? (3) **RQ3**: How does the performance of **SuperG** vary with different values of the hyper-parameters?

3.1 Experimental Setup

Datasets. We conduct experiments on three widely used real-world datasets: (a) Amazon Baby [22, 45], (b) Amazon Clothing, Shoes, and Jewelry [22, 45], and (c) TikTok [3]. For simplicity, we refer to them as Baby, Clothes, and TikTok. Both Baby and Clothes contain visual and textual modalities. TikTok is collected from the TikTok platform, recording users' viewed short videos. Its multi-modal features include visual, acoustic, and title textual information. The statistics of the pre-processed datasets are summarized in Appendix E. To simulate new items, we randomly select 20% of items and remove their historical interactions from the training process. Among these, half are further used as a validation set and the remaining half as a test set. These items are completely unseen during training.

Evaluation Metrics. We select two metrics that are widely used in personalized recommender systems: Recall (Recall@K) and Normalized Discounted Cumulative Gain (NDCG@K) [30]. Higher values of these two metrics indicate better model performance.

Baselines. To verify the effectiveness of SuperG, we compare it with multiple representative and state-of-the-art models that are suitable for the task scenarios: **MMGCN** [64], **LATTICE** [75], **MICRO** [76], **CCFCRec** [84], **GoRec** [2], **MILK** [3], **LGMRec** [20], **PROMPTMM** [62], **LLMTreeRec** [78], **MMHCL** [19], **MLLM-MSR** [71].

Implementation Details. We randomly split the user–item interaction data into training, validation, and test sets with a ratio of 8:1:1, following [3, 19]. The batch size is set to 1024. We adopt the Adam optimizer [34] with a learning rate tuned in $1e-3$, $1e-4$, $1e-5$ and select $1e-3$ as default. An early stopping strategy is applied to prevent overfitting. For model hyper-parameters, the embedding dimension d is set to 768, the number of retrieved similar users m is set to 5, and the number of retrieved behaviors per user k is also set to 5. The transformer encoder $\text{Trm}(\cdot)$ in Eq.(1) is implemented with 1 layer and 2 attention heads. The temperature parameter τ_1 in Eq.(2) is fixed at 0.1. The retrieval score weighting parameter λ in Eq.(6) is set to 0.6. The number of top behaviors N used to guide recommendation is set to 5. For the total loss, the weights are empirically set as $\alpha = 0.5$ and $\beta = 0.3$. We employ Llama-3-8B-Instruct as the backbone LLM, with the temperature parameter fixed at 0. We do not fine-tune the LLM due to the high computational cost and potential risk of user privacy leakage. For evaluation, we use the same prompt template as in training and set the maximum sequence length to 2048. All experiments are conducted five times with different random seeds, and we report the average results.

Table 1: Experimental results on Baby, Clothes, and TikTok datasets.

Models	Baby		Clothes		TikTok	
	Recall@10	Recall@20	Recall@10	Recall@20	Recall@10	Recall@20
MMGCN	0.0208	0.0278	0.0175	0.0289	0.1318	0.1385
LATTICE	0.0271	0.0328	0.0324	0.0371	0.1568	0.1695
MICRO	0.0283	0.0450	0.0340	0.0509	0.1760	0.1867
CCFCRec	0.0295	0.0464	0.0355	0.0545	0.1802	0.1920
GoRec	0.0335	0.0544	0.0429	0.0639	0.1465	0.1605
MILK	0.0381	0.0628	0.0484	0.0716	0.1886	0.2002
LGMRec	0.0409	0.0785	0.0552	0.0865	0.1956	0.2091
PROMPTMM	0.0653	0.0896	0.0681	0.0903	0.2029	0.2156
LLMTreeRec	0.0756	0.1108	0.0826	0.1153	0.2325	0.2561
MMHCL	0.0646	0.0950	0.0893	0.0925	0.2108	0.2389
MLLM-MSR	0.0859	0.1305	0.0906	0.1389	0.2302	0.2565
SuperG-SUEM	0.0892	0.1390	0.0951	0.1438	0.2571	0.2683
SuperG-IBMM	0.0906	0.1439	0.0989	0.1470	0.2596	0.2720
SuperG-BHRM	<u>0.0920</u>	<u>0.1477</u>	<u>0.1016</u>	<u>0.1508</u>	<u>0.2617</u>	<u>0.2766</u>
SuperG	0.0986	0.1507	0.1059	0.1562	0.2680	0.2812
	NDCG@10	NDCG@20	NDCG@10	NDCG@20	NDCG@10	NDCG@20
MMGCN	0.0138	0.0169	0.0165	0.0230	0.1201	0.1320
LATTICE	0.0151	0.0188	0.0178	0.0235	0.1536	0.1625
MICRO	0.0153	0.0192	0.0186	0.0243	0.1270	0.1357
CCFCRec	0.0159	0.0209	0.0207	0.0259	0.1799	0.1802
GoRec	0.0179	0.0249	0.0234	0.0292	0.1096	0.1124
MILK	0.0215	0.0277	0.0279	0.0343	0.1830	0.1852
LGMRec	0.0227	0.0283	0.0302	0.0455	0.1986	0.2091
PROMPTMM	0.0240	0.0298	0.0331	0.0476	0.2029	0.2104
LLMTreeRec	0.0345	0.0410	0.0415	0.0553	0.2520	0.2685
MMHCL	0.0336	0.0401	0.0389	0.0506	0.2509	0.2668
MLLM-MSR	0.0457	0.0526	0.0491	0.0568	0.2607	0.2692
SuperG-SUEM	0.0551	0.0613	0.0533	0.0690	0.2755	0.2807
SuperG-IBMM	0.0590	0.0681	0.0585	0.0749	0.2786	0.2862
SuperG-BHRM	<u>0.0603</u>	<u>0.0706</u>	<u>0.0602</u>	<u>0.0772</u>	<u>0.2802</u>	<u>0.2883</u>
SuperG	0.0619	0.0720	0.0625	0.0792	0.2836	0.2912

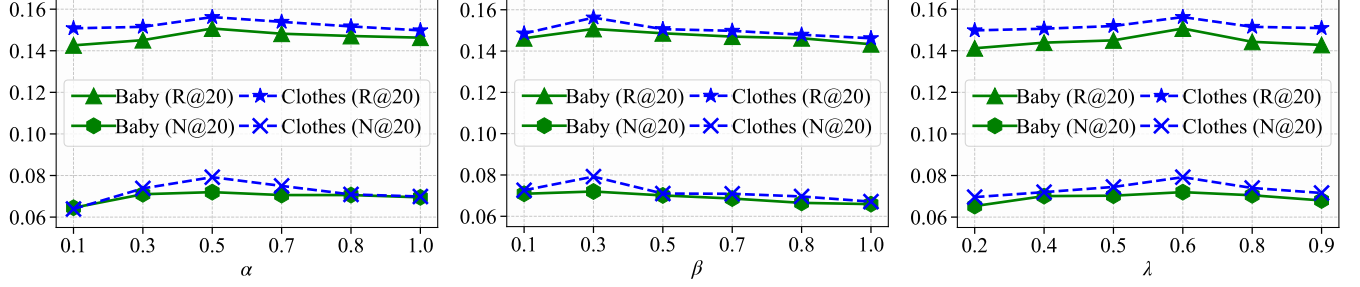
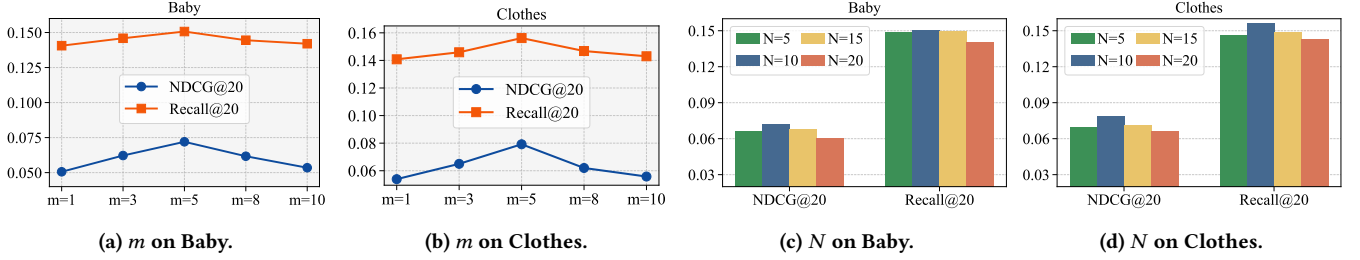
3.2 Overall Comparisons (RQ1)

As shown in Table 2, we compare our model with competitive baselines on three datasets, and obtain the following observations: (1) On all three datasets, SuperG consistently outperforms all baselines. Specifically, it improves the strongest baseline (MLLM-MSR) in terms of Recall@10 by 14.8%, 16.9%, and 16.4% on the Baby, Clothes, and TikTok datasets, respectively. These results demonstrate the effectiveness of our framework, and we attribute the gains to the three proposed modules that jointly enhance the performance of the model. (2) Compared with representative baselines such as MMHCL, LGMRec, CCFCRec, and MMGCN, SuperG achieves consistently better performance. For example, on the Clothes dataset, SuperG improves over the baseline MMHCL by 18.6% in Recall@10. We attribute these gains to the similar user exploration module, which avoids spurious neighbors caused by relying solely on simple co-occurring interactions and instead introduces reliable collaborative signals based on explicit user–user similarity. (3) Compared with recent LLM-based models such as LLMTreeRec, PROMPTMM, and MLLM-MSR, SuperG also demonstrates clear superiority. On the TikTok dataset, for instance, SuperG improves over the strongest baseline (MLLM-MSR) by 16.4% in Recall@10. This improvement

stems from the informative behavior mining module, which explicitly retrieves and reranks the most informative interactions from both the target user and similar users.

3.3 Ablation Study (RQ2)

To exploit the effectiveness of each component in the proposed **SuperG**, we conduct ablation studies on multiple datasets. As illustrated in Table 1, we compare the full model with several variants: **SuperG-SUEM**: the Similar User Exploration Module (SUEM) is removed, and informative behaviors are directly retrieved from all users’ historical interactions. **SuperG-IBMM**: the Informative Behavior Mining Module (IBMM) is removed. In this setting, we retrieve user behaviors without incorporating LLM feedback. **SuperG-BHRM**: the Behavior-Guided Hybrid Recommendation Module (BHRM) is removed, and external embeddings are directly appended to the final prompt without LLM-compatible representation adaptation. From the results, we derive the following observations: (1) Each component contributes substantially to the superior performance of SuperG. SUEM contributes the most to performance, followed by IBMM and BHRM. For example, on the Baby dataset, removing SUEM, IBMM, and BHRM leads to performance drops of 12.3%, 4.9%, and 2.7% in NDCG@10, respectively, underscoring their indispensable roles. (2) SuperG-SUEM underperforms the full

Figure 4: The experimental results on varying α , β and λ .Figure 5: The experimental results on varying m and N .

model, demonstrating that merely retrieving behaviors from all users cannot effectively capture informative collaborative signals. This validates that the proposed SUEM is critical for tackling CH1, as it enables precise identification of similar users while avoiding spurious neighbors. (3) Both SuperG-IBMM and SuperG-BHRM perform significantly worse than SuperG, highlighting the importance of explicitly retrieving informative behaviors and adapting user-item embeddings into LLM-compatible representations. These results show that IBMM and BHRM can effectively identify the most informative behaviors and leverage these behaviors to guide the final recommendation for tackling CH2.

3.4 Effect of Hyper-Parameters (RQ3)

We further investigate the effects of key hyper-parameters on the performance of SuperG on the Baby and Clothes datasets.

Effect of the number of similar users m . The parameter m controls the number of similar users retrieved in the similar user exploration module. As shown in Fig.5(a)-(b), a small m (e.g., $m = 1$) leads to insufficient collaborative signals, while a very large m introduces noisy and spurious neighbors. The best performance is achieved at $m = 5$, which we adopt as the default setting.

Effect of the retrieval score weighting parameter λ . We vary λ in 0.2, 0.4, 0.5, 0.6, 0.8, 1.0, and report Recall@20 and NDCG@20 in Fig.4. When λ is small (e.g., 0.2), the score relies heavily on semantic relevance while largely ignoring user-specific preferences, resulting in suboptimal performance. Conversely, a large λ (e.g., 1.0) overly emphasizes user-specific preferences and disregards behavioral similarity, leading to performance degradation. The best balance is observed at $\lambda = 0.6$, which we set as the default.

Effect of top- N behaviors for guiding recommendation. As illustrated in Fig.5(c)-(d), too few behaviors (e.g., $N = 5$) fail to provide sufficient informative signals, while too many behaviors introduce redundant or noisy interactions. Performance peaks at $N = 10$, and we therefore set $N = 10$ as the default.

Effect of the total loss weights α and β . We conduct sensitivity analysis by varying one parameter while fixing the other. As shown in Fig.4, model performance remains relatively stable within a reasonable range, demonstrating the robustness of SuperG. Empirically, we set $\alpha = 0.5$ and $\beta = 0.3$, which provides the best balance among the loss objectives.

4 Conclusion

In this paper, we propose the Joint Similar User Exploration and Informative Behavior Guidance model (SuperG) to address the Multi-Modal New Item Recommendation (MMNIR) problem. SuperG consists of three modules: the similar user exploration module, the informative behavior mining module, and the behavior-guided hybrid recommendation module. The similar user exploration module identifies users with preferences similar to the target user via user representation contrastive learning. The informative behavior mining module retrieves the most informative behaviors from both the target user and similar users' histories through LLM-based feedback. The behavior-guided hybrid recommendation module incorporates these behaviors to guide the LLM in generating the final recommendation. By integrating these three modules, SuperG effectively leverages rich information from informative behaviors to enhance the recommendation of new multi-modal items. We also conduct extensive experiments on multiple datasets to demonstrate the superior performance of our proposed SuperG model.

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Appendix

A Related Work

Multi-Modal Recommendation. Recent research in MMR leverages multi-modal item content such as text, images, and audio to

improve recommendation quality [49, 58, 69, 83]. Among these methods, GNNs have been widely adopted to propagate embeddings over user-item interaction graphs [25, 60, 65] enriched with multi-modal features, effectively capturing user preferences across different content types [8, 23, 35, 39]. Although these methods have made notable progress in utilizing heterogeneous modality signals, most of them heavily depend on CF information [67]. This reliance limits their generalization capabilities, particularly in handling newly emerging items that lack sufficient interaction history.

Multi-Modal New Item Recommendation. To alleviate the limitations of MMR approaches in cold-start scenarios, recent studies explore models to recommend newly introduced items by leveraging their multi-modal content representations [18, 37, 76]. These approaches can be broadly categorized into two types: content-based methods and hybrid recommendation methods. Content-based methods exclusively rely on modeling item characteristics extracted from multi-modal inputs, enabling recommendation in the absence of new item interactions. Hybrid recommendation methods integrate both collaborative filtering signals and multimedia features during training to construct joint user-item representations that encode hybrid preference signals [7, 59, 84]. Despite their effectiveness, such methods typically assume that the data is consistently complete and of high quality. The absence of modalities either during model training or inference significantly degrades recommendation performance, highlighting the need for methods that are robust to modality incompleteness. To address the challenge of modality missingness in MMNIR, several methods propose different strategies [3, 16, 74]. MILK [3] employs invariant learning [9, 10, 81] to maintain robustness under incomplete modality scenarios and adopts a Dirichlet-based cyclic mixup strategy to enhance modality representation balance. SiBraR [16] uses a single extractor with shared weights to map multiple modalities into a common representation space, aiming to ensure robustness against missing modalities. However, MILK relies on sensitive parameter choices in its Dirichlet-based mixup strategy [4, 77, 86], which may hinder the balance of modality representations and affect generalization. SiBraR adopts a tightly coupled architecture where the absence of any modality impacts the shared extractor, thereby degrading the extraction quality of the remaining modalities.

B User Representation Construction

Item Representation. For each item $j \in \mathcal{V}$, we extract modality-specific features from three modalities: **visual**, **acoustic**, and **textual**.

Textual modality. We employ the pretrained RoBERTa-base model to encode the textual description of each item. The hidden representation of the [CLS] token from the final layer is used as the textual feature vector, $\mathbf{x}_j^{(txt)} \in \mathbb{R}^{768}$.

Visual modality. We adopt BLIP to extract visual features from item images. The global embedding output from the image encoder serves as the visual feature vector, $\mathbf{x}_j^{(vis)} \in \mathbb{R}^{768}$.

Acoustic modality. For audio signals associated with items, we use BEATs. Frame-level features from the final transformer layer are averaged to form a global acoustic embedding, $\mathbf{x}_j^{(aud)} \in \mathbb{R}^{768}$.

Thus, for each item j , we obtain three modality-specific embeddings:

$$\mathbf{x}_j = \{\mathbf{x}_j^{(vis)}, \mathbf{x}_j^{(aud)}, \mathbf{x}_j^{(txt)}\}.$$

To obtain a unified item representation $\mathbf{z}_j$ with the same dimensionality as each modality, we compute the mean of the modality-specific embeddings:

$$\mathbf{z}_j = \frac{1}{3}(\mathbf{x}_j^{(vis)} + \mathbf{x}_j^{(aud)} + \mathbf{x}_j^{(txt)}), \quad \mathbf{z}_j \in \mathbb{R}^{768}.$$

This simple mean-pooling operation ensures that all modalities contribute equally to the final representation while avoiding the introduction of additional trainable parameters. The resulting $\mathbf{z}_j$ can then be used for subsequent sequence modeling and recommendation tasks. For the TikTok dataset, all three modalities (visual, acoustic, and textual) are available for each item. For the Baby and Clothes datasets, only the visual and textual modalities are present, and the acoustic modality is not used. In these cases, the mean-pooling is performed over the available modalities only.

C Behavior-Guided Hybrid Recommendation Module

Prompt Construction Details. In this section, we provide a detailed illustration of the prompt design used in our framework. The prompt consists of four main parts: (1) target user information, (2) new item information, (3) informative behaviors, and (4) instruction. The target user information, the new item information, and the informative behaviors all include textual descriptions and embeddings.

Target User. The target user is represented by a unique ID, textual attributes (when available), and the corresponding embedding.

New Item. The candidate item is represented by its ID, textual metadata (e.g., title, category), and its embedding.

Informative Behaviors. We convert the retrieved informative behaviors into textual form, where each behavior describes whether a specific user interacted with a specific item. Both IDs and embeddings are included.

Instruction. Finally, we prepend an instruction that specifies the prediction task and the expected output format:

Instruction:

"Based on the above information, predict whether the target user will interact with the new item. Respond with 'Yes' or 'No'."

As illustrated in Fig. 6, the complete prompt is formed by concatenating the instruction, target user information, new item information, and informative behaviors. This design allows the LLM to simultaneously leverage structured metadata, textual descriptions, and dense embeddings for personalized recommendation.

D Inference

Inference Algorithm of SuperG. The inference procedure is detailed in Algorithm 1. During inference, SuperG takes as input a target user and a new item, and outputs a binary prediction (Yes/No) indicating whether the user would interact with the item. The inference process consists of four steps: **1. Similar User Retrieval.** We first compute the embedding of the target user using Eq.(1). By measuring cosine similarities with all users, we retrieve the top- m most similar users. These users provide collaborative signals to compensate for the cold-start challenge in new item recommendation. **2. Informative Behavior Retrieval.** From the histories of the

Algorithm 1 Inference Procedure of SuperG

Require: Target user u , new item j , trained SuperG model, parameters m, k, N

Ensure: Prediction $\hat{y}_{u,j} \in \{\text{Yes}, \text{No}\}$

- 1: **Step 1: Similar User Retrieval**
- 2: Obtain user embedding $\mathbf{e}_u$ by Eq.(1)
- 3: Compute cosine similarity between $\mathbf{e}_u$ and all other users' embeddings
- 4: Retrieve top- m most similar users $\{u_1, \dots, u_m\}$
- 5: **Step 2: Informative Behavior Retrieval**
- 6: For each similar user u_l and target user u , retrieve top- k historical interactions based on Eq.(6)
- 7: Form candidate behavior set $\mathcal{B}^{\text{retrieved}}$ with size $(m+1) \times k$
- 8: Rank all candidates and select top- N informative behaviors $\mathcal{B}_{\text{ranked}}$
- 9: **Step 3: Prompt Construction**
- 10: Convert $\mathcal{B}_{\text{ranked}}$ into textual descriptions
- 11: Project embeddings $\mathbf{e}_u, \mathbf{z}_j$ into LLM space via adapter $f(\cdot)$
- 12: Construct final prompt = {instruction, target user & new item info, informative behaviors}
- 13: **Step 4: Interaction Prediction**
- 14: Feed prompt into LLM
- 15: Obtain binary prediction $\hat{y}_{u,j} \in \{\text{Yes}, \text{No}\}$

Target Behavior:

Target User: ID + Attributes + [USER_EMBED]

New Item: ID + Attributes + [ITEM_EMBED]

Informative Behaviors:

Behavior 1:

User l_1 : ID + Attributes + [USER_EMBED]

Item r_1 : ID + Attributes + [ITEM_EMBED]

User l_1 interacted with Item r_1 ? "Yes"

Behavior 2:

User l_2 : ID + Attributes + [USER_EMBED]

Item r_2 : ID + Attributes + [ITEM_EMBED]

User l_2 interacted with Item r_2 ? "No"

.....

Instruction:

Based on the above information, predict whether the target user will interact with the new item.
Respond with 'Yes' or 'No'.

target user and similar users, we first retrieve the top- k historical interactions based on the retrieval score $S_{u,j,l,r}^{\text{retriever}}$ (Eq.(6)) to form the candidate set. We then rerank the candidates based on $S_{u,j,l,r}^{\text{retriever}}$, and finally select the top- N informative behaviors $\mathcal{B}_{\text{ranked}}$. These behaviors are the most relevant and user-personalized signals for supporting prediction. **3. Prompt Construction.** The selected behaviors are transformed into textual descriptions. Meanwhile, the embeddings of the target user and new item are projected into the LLM's semantic space via the adapter module $f(\cdot)$. We then construct a unified prompt that combines (i) an instruction specifying the prediction task, (ii) the target user and new item information, and (iii) the informative behaviors. **4. Interaction Prediction.** The constructed prompt is fed into the LLM, which outputs a binary response ("Yes" or "No") representing the prediction result. Through these steps, SuperG effectively leverages both collaborative information from similar users and supportive signals from informative behaviors to improve recommendation performance in MMNIR scenarios.

E Experiment

Table 2: The statistics of datasets.

Dataset	Baby	Clothes	TikTok
Train - # Users	19,442	39,384	9,319
Train - # Items	5,640	18,427	5,368
Train - # Interactions	128,963	222,759	55,126
Train - Density (%)	0.118	0.031	0.110
Validation - # Users	10,342	19,801	2,380
Validation - # Items	705	2,303	671
Test - # Users	10,474	19,858	2,960
Test - # Items	705	2,303	671

E.1 Experimental Setup

Datasets. We conduct experiments on three widely used real-world datasets: (a) Amazon Baby [22, 45], (b) Amazon Clothing, Shoes, and Jewelry [22, 45], and (c) TikTok [3]. For simplicity, we refer to them as Baby, Clothes, and TikTok. Both Baby and Clothes contain visual and textual modalities. TikTok is collected from the TikTok platform, recording users' viewed short videos. Its multi-modal features include visual, acoustic, and title textual information. The statistics of the pre-processed datasets are summarized in Table 2. To simulate new items, we randomly select 20% of items and remove their historical interactions from the training process. Among these, half are further used as a validation set and the remaining half as a test set. These items are completely unseen during training.

Figure 6: Illustration of the prompt in SuperG.